

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1489

Greenbrae 138 kV
June 20, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1489. The Interconnection Customer proposes a 52.8 MW solar photovoltaic facility interconnecting at the Greenbrae 138 kV substation in the Virginia area. The request was received on June 05, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 372 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,699,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
MIDWAY SUBSTATION - VIRGINIA TECH POWER PLANT 138kV	E PORTAL - PINNACLE PEAK SRP 138kV	119.1	218	62.2
SIMPSON - HINMAN 138kV	HINMAN - FLOYD 138kV	118.4	221	100.0
HIGHWAY 59 - W BASSETT SUBSTATION 138kV	HINMAN - FLOYD 138kV	109.7	220	100.0
SIMPSON - PINNACLE PEAK SRP 138kV L	HINMAN - FLOYD 138kV	107.7	230	100.0
MIDWY - S CHRISTIANSBURG 138kV	EAST PORTAL - PINNACLE PEAK SRP 138kV	106.2	281	62.2
SOUTH CHRISTIANSBURG - HAPPY CANYON 138kV	PINNACLE PEAK SRP - E PORTAL 138kV L	105.7	309	62.2
HAPPY CANYON - EDGEMONT 138kV	E PORTAL SUBSTATION - PINNACLE PEAK SRP 138kV	105.3	336	62.2
SHULLSURG - EDGEMONT 138kV	PINNACLE PEAK SRP - EAST PORTAL 138kV	104.9	363	62.2
RILEY - HINMAN 138kV	BEAVER - EUFAULA SUBSTATION 500kV	104.9	261	100.0
WEST COOKEVILLE - RILEY 138kV	CARTON HILLS - CLINCH RIV 345kV	104.8	269	100.0
GREENBRAE - WEST COOKEVILLE 138kV	CLINCH RIVER - CARLTON HILLS SUBSTATION 345kV Line	104.5	289	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
VIRGINIA TECH POWER PLANT - MIDWAY 138kV	Rebuild line with higher-capacity conductor	\$8,301,000	10% (\$837,000)
HINMAN - SIMPSON 138kV	Rebuild line with higher-capacity conductor	\$48,022,000	20% (\$9,763,000)
HIGHWAY 59 - W BASSETT SUBSTATION 138kV	Reconductor line	\$4,322,000	20% (\$879,000)
SIMPSON - PINNACLE PEAK SRP 138kV	Reconductor line	\$2,726,000	20% (\$554,000)
MIDWAY - SOUTH CHRISTIANSBURG 138kV	Reconductor line	\$2,550,000	10% (\$257,000)
SOUTH CHRISTIANSBURG - HAPPY CANYON 138kV	Reconductor line	\$1,103,000	10% (\$111,000)
HAPPY CANYON - EDGEMONT 138kV	Reconductor line	\$415,000	10% (\$42,000)
SHULLSURG - EDGEMONT 138kV	Reconductor line	\$1,470,000	10% (\$148,000)
RILEY - HINMAN 138kV	Reconductor line	\$2,687,000	20% (\$546,000)
WEST COOKEVILLE - RILEY 138kV	Reconductor line	\$3,670,000	20% (\$746,000)
W COOKEVILLE - GREENBRAE 138kV	Reconductor line	\$4,014,000	20% (\$816,000)
	Total allocated to project		\$14,699,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.